MINI PROJECT - Global Video Game Sales Analysis

AIM:

To clean, analyze, and visualize global video game sales data using Python and Power BI to identify sales trends over time, the best-performing genres, platforms, and publishers, and the differences in buying behaviour across North America, Europe, Japan, and the rest of the world.

DOMAIN:

Gaming Industry Analytics / Sales Analytics / Business Intelligence.

PROBLEM STATEMENT:

The video game industry spans dozens of hardware platforms, hundreds of publishers, and very different regional markets. Publishers and retailers need to understand which genres and platforms drive revenue, how sales have changed over the years, and how player preferences differ by region. This project analyzes historical video game sales data using Python and Power BI to generate meaningful insights and support data-driven decisions.

DATASET DESCRIPTION:

The project uses the public **Video Game Sales (vgsales)** dataset, which lists every game that sold more than 100,000 copies. It contains **16,598 records** and **11 fields**, covering 31 platforms, 12 genres, and 577 publishers from 1980 to 2020. Sales figures are in millions of units.

- **Rank** - overall rank by global sales
- **Name** - game title
- **Platform** - console or system (PS2, Wii, X360, DS, etc.)
- **Year** - year of release
- **Genre** - game genre (Action, Sports, Shooter, etc.)
- **Publisher** - publishing company
- **NA_Sales** - North America sales
- **EU_Sales** - Europe sales
- **JP_Sales** - Japan sales
- **Other_Sales** - rest of the world sales
- **Global_Sales** - total worldwide sales

PROCEDURE:

1. Loaded the vgsales dataset using Python and Pandas.
2. Inspected dataset structure, dimensions, data types, missing values, and duplicate records.
3. Cleaned and standardized text fields (Name, Platform, Genre, Publisher).
4. Handled missing Year and Publisher values.
5. Removed duplicate game records.
6. Validated that regional sales add up to global sales.
7. Created derived features such as regional share, dominant region, decade, and sales category.
8. Performed exploratory data analysis using Pandas, Matplotlib, and Seaborn.
9. Analyzed sales by year, genre, platform, publisher, and region.
10. Compared genre preferences across regional markets.
11. Created static Python visualizations.
12. Imported the data into Power BI and built calculated measures for target sales.
13. Created an interactive Video Game Sales Dashboard in Power BI.

KEY INSIGHTS:

The analysis will investigate:

- Total global sales and the contribution of each region.
- Global sales trend across release years and genres.
- Genres with the highest global sales.
- Genres with the highest sales in each region (NA, EU, JP, Other).
- Platforms with the highest global sales.
- Platforms with the highest number of titles.
- Publishers with the highest sales.
- Best-selling individual games.
- Differences in genre preference between regions.
- Correlation between regional and global sales.
- Actual global sales compared with the target.
- Distribution of titles by sales category.

PYTHON ANALYSIS

Python was used for data cleaning, processing, exploratory data analysis, feature engineering, and visualization.

The main libraries used were:

- Pandas
- NumPy
- Matplotlib
- Seaborn

DATA CLEANING

The dataset was loaded using Pandas. The following operations were performed:

- Checked dataset structure and dimensions.
- Examined data types.
- Identified missing values - 271 records had no Year and 58 records had no Publisher.
- Standardized text values by removing unnecessary spaces.
- Converted Year into a numeric (integer) field.
- Filled missing Publisher values with "Unknown".
- Kept records with a missing Year for overall totals, but excluded them from year-wise analysis.
- Identified and removed 2 duplicate records (same game, platform, and year).
- Checked sales fields for zero and negative values (none were found).
- Validated that NA + EU + JP + Other sales match Global_Sales (differences are within 0.02M rounding).
- Noted that 2017 and 2020 contain only a few records, so these years are incomplete.

The dataset contained **16,596 records** after cleaning.

DATA PROCESSING / FEATURE ENGINEERING

1. Regional Sales Share

Regional Share = Regional Sales / Global Sales

This shows how much of each game's sales came from North America, Europe, Japan, and other regions.

2. Dominant Region

Dominant Region = Region with the highest sales for the game

This identifies the main market for each title.

3. Decade

Decade = (Year // 10) x 10

Groups games into the 1980s, 1990s, 2000s, and 2010s for long-term trend comparison.

4. Sales Category

Niche (<0.1M) | *Moderate (0.1-1M)* | *Hit (1-5M)* | *Blockbuster (>5M)*

Games were grouped into four categories based on global sales to support filtering and comparison.

5. Target Sales (Power BI)

Target Sales = calculated goal measure in Power BI

A target measure was created in Power BI and compared with actual global sales using a KPI visual.

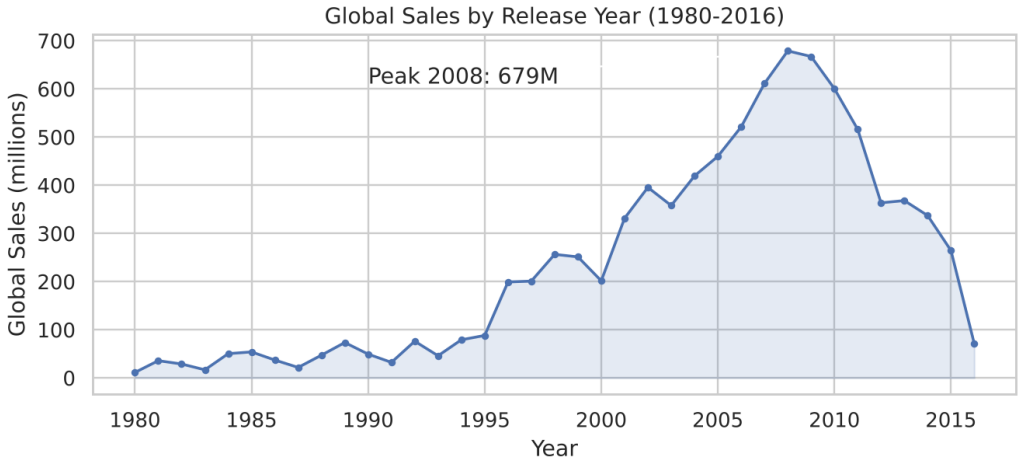
EXPLORATORY DATA ANALYSIS

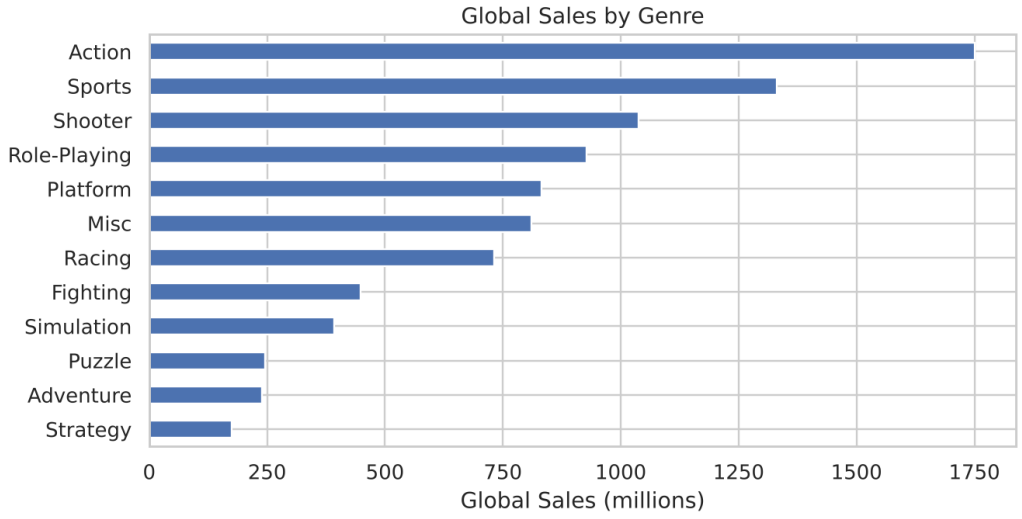
Python will be used to analyze:

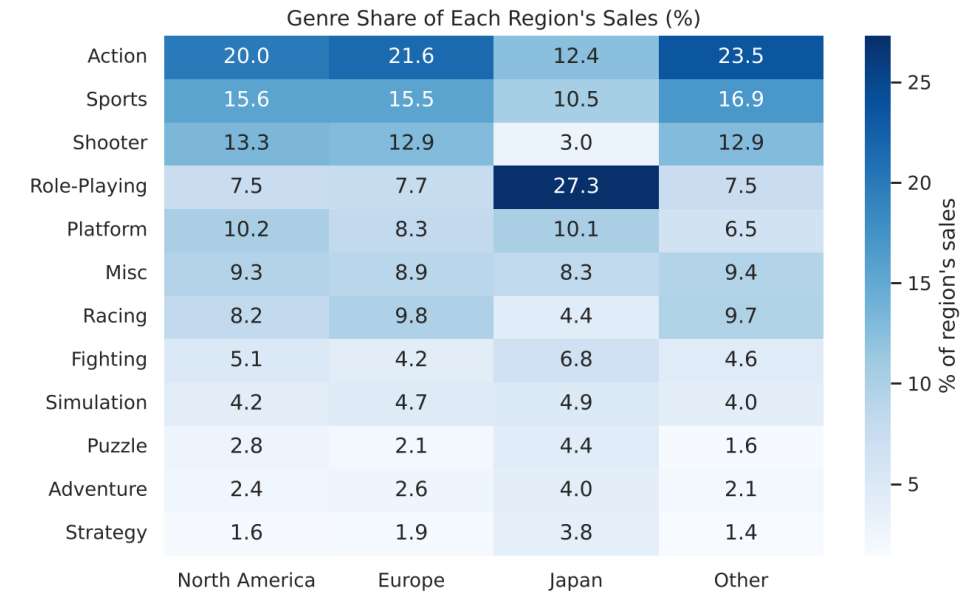
- Global sales by release year
- Global sales by genre
- Regional sales by genre
- Genre share of each region's sales
- Global sales by platform
- Number of titles by platform
- Top publishers by global sales
- Top 10 best-selling games
- Share of sales by region
- Correlation between regional and global sales
- Number of titles by sales category

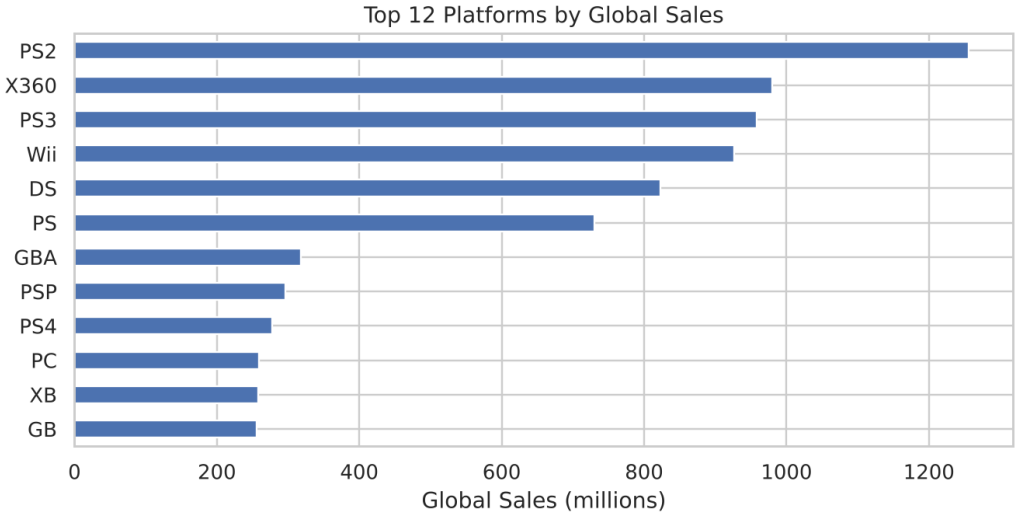
PYTHON VISUALIZATIONS

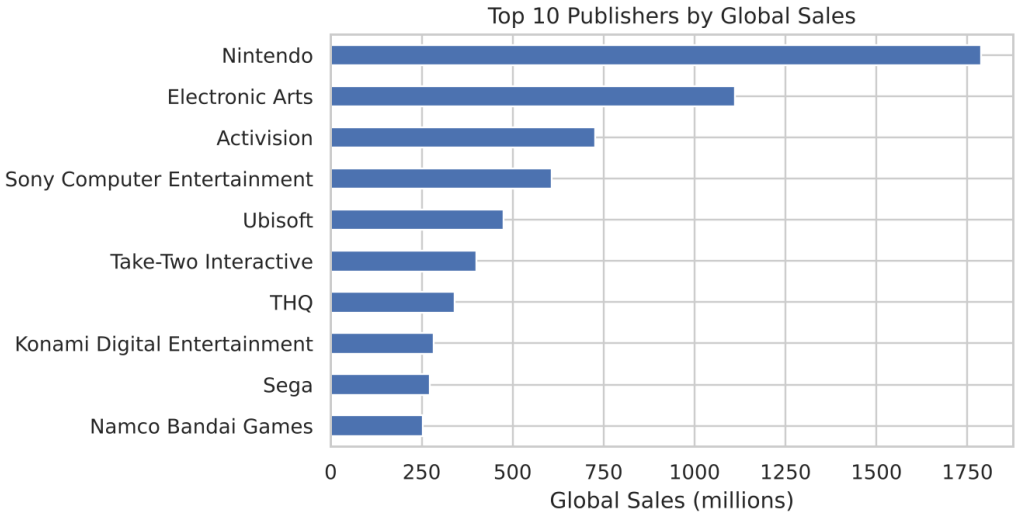
Python visualizations were created using Matplotlib and Seaborn to analyze yearly trends, genres, platforms, publishers, best-selling titles, and regional sales patterns. The results were used to guide the design of the Power BI dashboard.

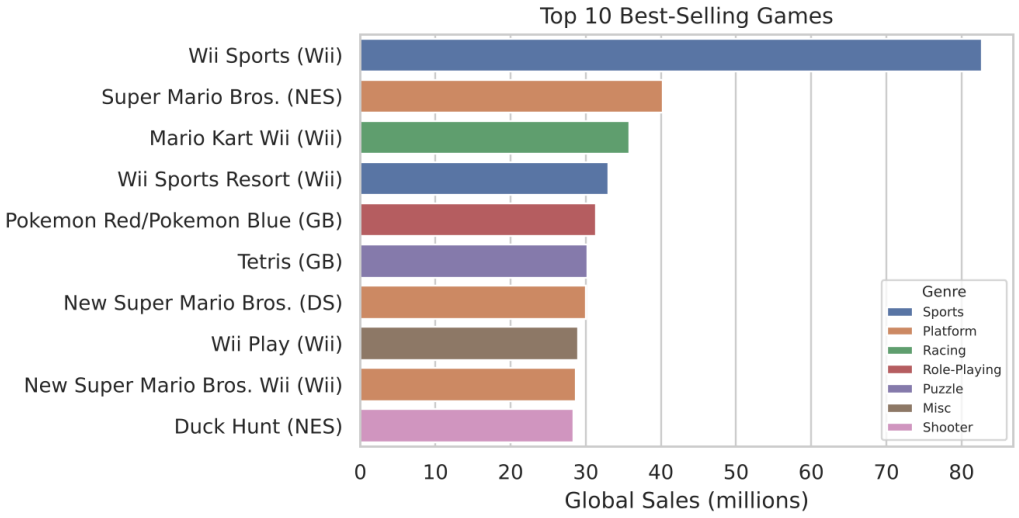


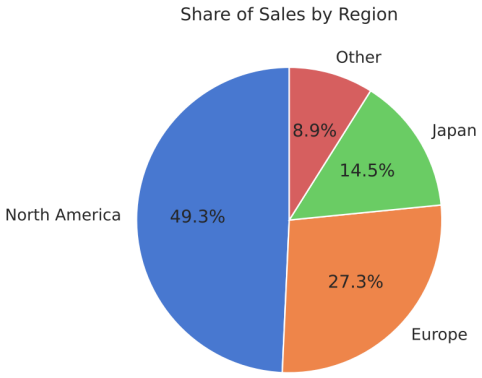


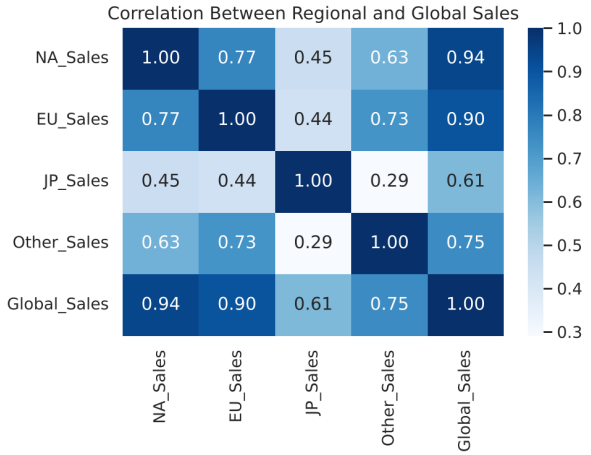

Genre Share of Each Region's Sales (%)
Action 20.0 21.6 12.4 23.5
Sports 15.6 15.5 10.5 16.9
Shooter 13.3 12.9 3.0 12.9
Role-Playing 7.5 7.7 27.3 7.5
Platform 10.2 8.3 10.1 6.5
Misc 9.3 8.9 8.3 9.4
Racing 8.2 9.8 4.4 9.7
Fighting 5.1 4.2 6.8 4.6
Simulation 4.2 4.7 4.9 4.0
Puzzle 2.8 2.1 4.4 1.6
Adventure 2.4 2.6 4.0 2.1
Strategy 1.6 1.9 3.8 1.4
North America
Europe
Japan
Other
% of region's sales
25
20
15
10
5

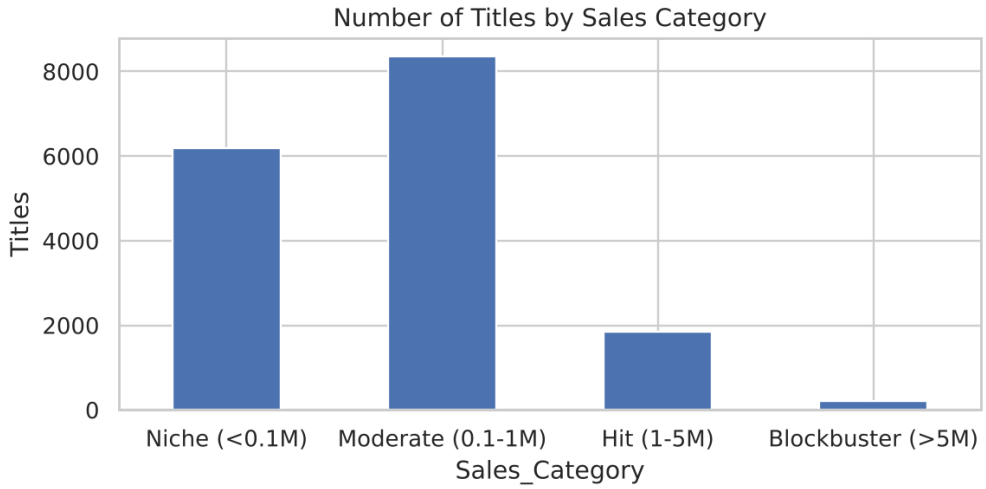

Top 12 Platforms by Global Sales
PS2
X360
PS3
Wii
DS
PS
GBA
PSP
PS4
PC
XB
GB
0
200
400
600
800
1000
1200
Global Sales (millions)


Top 10 Publishers by Global Sales
Nintendo
Electronic Arts
Activision
Sony Computer Entertainment
Ubisoft
Take-Two Interactive
THQ
Konami Digital Entertainment
Sega
Namco Bandai Games
0
250
500
750
1000
1250
1500
1750
Global Sales (millions)









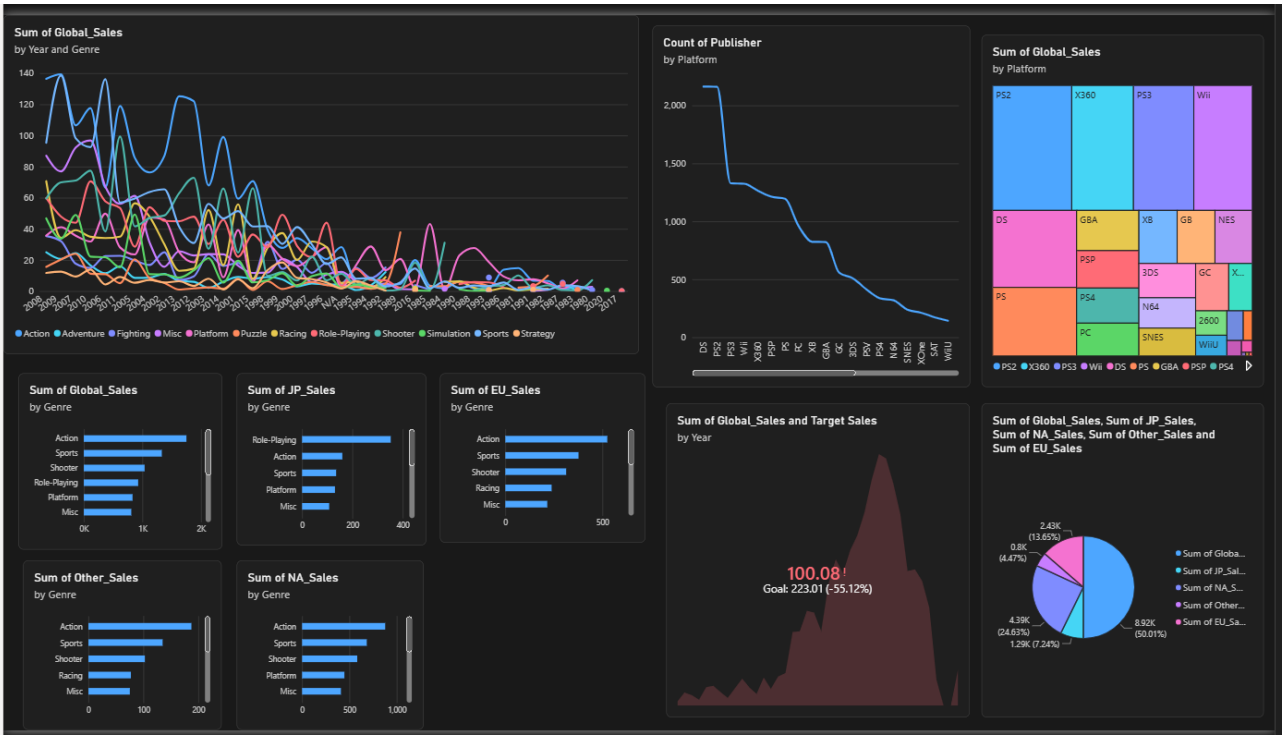
OBSERVATIONS:

- Total global sales are about **8,920 million** units. North America is the largest market (49.3% of regional sales), followed by Europe (27.3%), Japan (14.5%) and other regions (8.9%).
- Sales peaked in **2008** (679M units) and declined after 2011. Figures after 2015 are low partly because the dataset is incomplete for recent years.
- **Action** is the top genre (1,751M), followed by Sports (1,331M) and Shooter (1,037M).

- Japan behaves very differently: **Role-Playing** is its top genre (27.3% of Japanese sales), while Shooter games sell very little there.
- **PS2** is the best-selling platform (1,256M), followed by X360 and PS3. DS and PS2 have the largest number of titles (over 2,100 each).
- **Nintendo** leads publishers by sales (1,787M), while Electronic Arts has the most titles. Most of the top 10 best-selling games are Nintendo titles, led by Wii Sports (82.7M).
- North American sales have the strongest correlation with global sales (0.94); Japan has the weakest (0.61).
- Around 87% of titles sold under 1 million units - only a small number of blockbusters drive most of the industry's revenue.

Power BI Output

The cleaned dataset was imported into Power BI to create an interactive Video Game Sales Dashboard. The dashboard uses a line chart to show global sales by year and genre, bar charts for global, North America, Europe, Japan, and other-region sales by genre, a line chart showing the count of titles by platform, and a treemap showing global sales by platform. A KPI visual compares actual global sales against the target sales goal, and a pie chart breaks down total sales by region.



Dashboard Highlights:

- **Sum of Global_Sales by Year and Genre** - shows the rise in sales through the 2000s, with Action, Sports, and Shooter leading in most years.
- **Count of Publisher by Platform** - DS and PS2 have the largest number of titles (about 2,160 each), followed by PS3, Wii, and X360.
- **Sum of Global_Sales by Platform (treemap)** - PS2, X360, PS3, Wii, and DS account for the largest share of sales.
- **Sales by Genre (Global, NA, EU, JP, Other)** - Action is the top genre in every region except Japan, where Role-Playing leads.
- **Global Sales vs Target (KPI)** - the KPI shows actual sales of 100.08 against a goal of 223.01, which is 55.12% below target.
- **Regional Sales Breakdown (pie)** - Global 8.92K, NA 4.39K, EU 2.43K, JP 1.29K, and Other 0.80K (millions of units).

RESULT:

The project successfully transforms the raw video game sales dataset into a cleaned analytical dataset using Python and presents the resulting insights through an interactive Power BI dashboard. The project provides a comprehensive view of sales trends over time, the best-performing genres, platforms, and publishers, and the differences between regional markets. The visualizations help in understanding which genres and platforms drive sales, how the market grew and declined over the years, and how player preferences in Japan differ from those in North America and Europe.